

Configuring the STI Control Software

STI Control Software is a PC based program that controls the Ascension 200 mount. The motor controller may also be used on other mounts that use the Sidereal Technology motion controller, including mounts made by Mathis Instruments. STI also acts as an ASCOM driver for the mount. The main STI window is shown below.

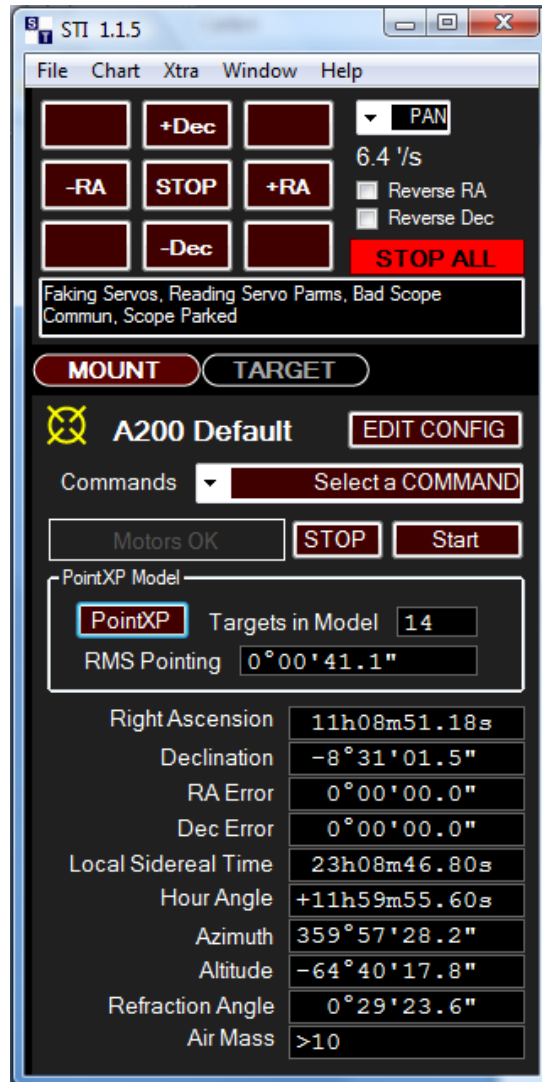


Figure 24 - STI home screen

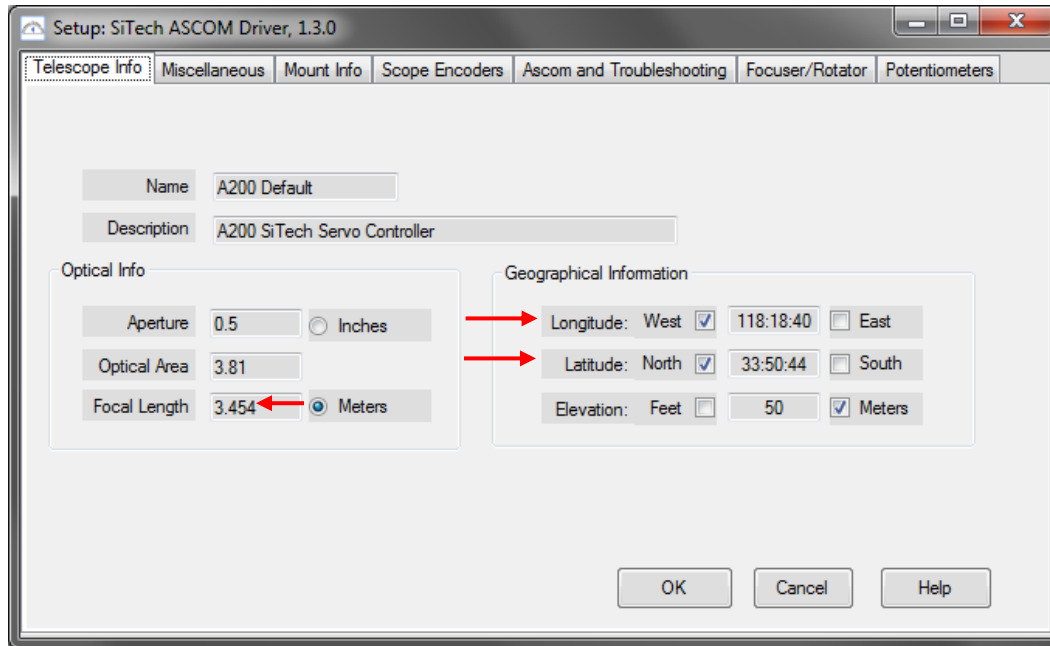
Begin by connecting the supplied USB cable to the USB port on the mount electronics box, and plugging the other end into the computer or a connected USB hub. Windows should detect the new device and load the proper drivers, which were installed with the STI installer.

Next, launch STI and click on the "EDIT CONFIG" button.



Figure 25 - Edit Config button

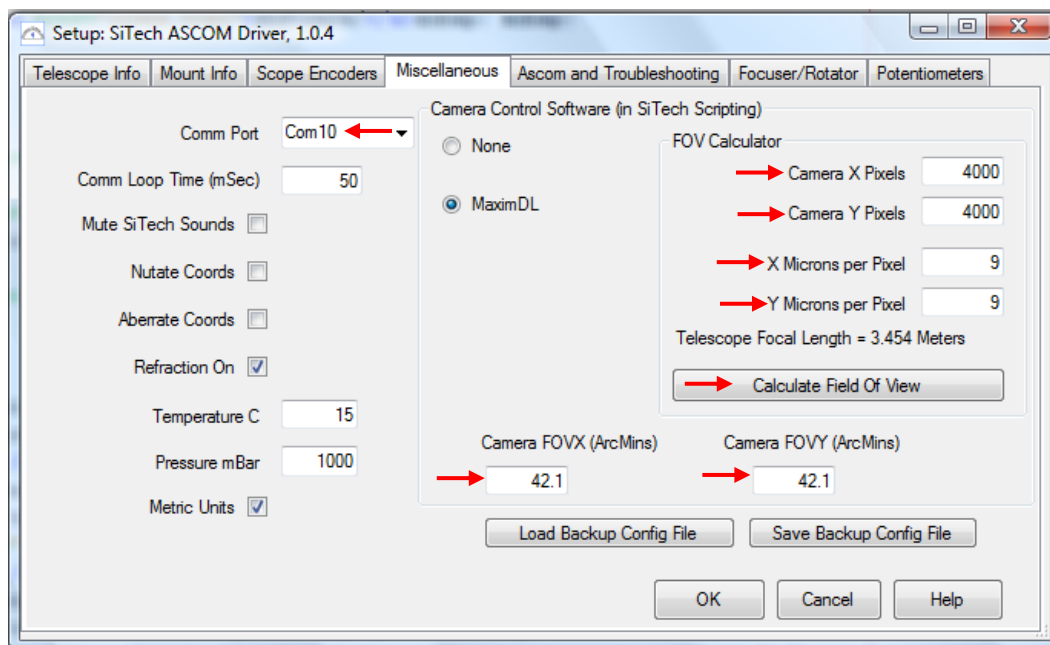
Telescope Info Tab



The first tab in the Setup screen contains some basic parameters about your site and telescope. The most important parameters to adjust on this screen are:

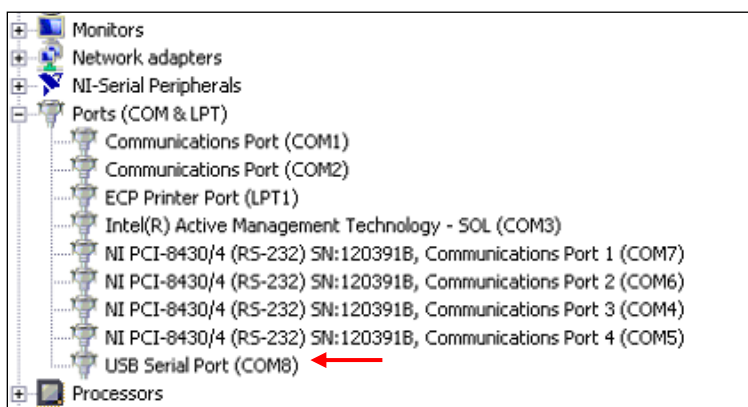
- Longitude and Latitude (the mount must know its location in order to accurately point at stars)
- Focal Length of the telescope (in order to reliably find PlateSolve solutions during a mount model.)

Miscellaneous Tab



The most important setting on this screen is the "Comm Port". You must set this to the COM port that is associated with the with USB connection on the mount electronics box.

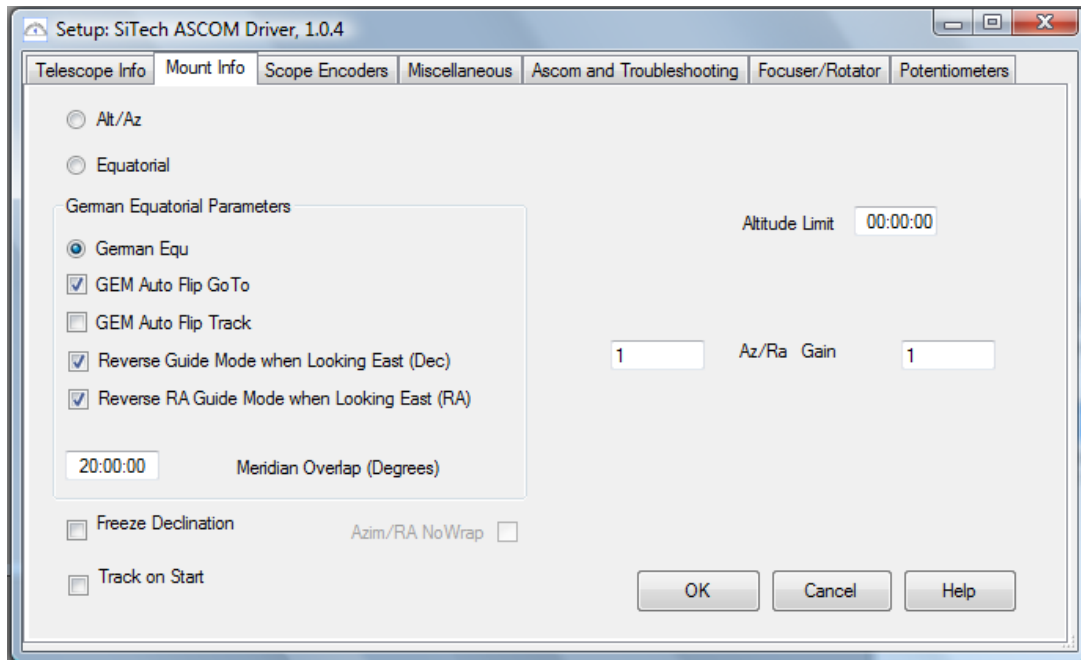
One method for identifying the correct COM port is to open Windows Device Manager, expand the "Ports" category, and look for the port that is listed as a "USB Serial Port". In the example below, the A200 mount is connected on COM8. Note however that some other devices, such as Boltwood II Cloud Sensors, may also be identified as USB Serial Ports.



In addition, the "Camera FOVX" and "Camera FOVY" values should be configured to the unbinned field of view for your camera / telescope combination. These numbers are used by PlateSolve during the automated mount modeling process to determine your approximate image scale.

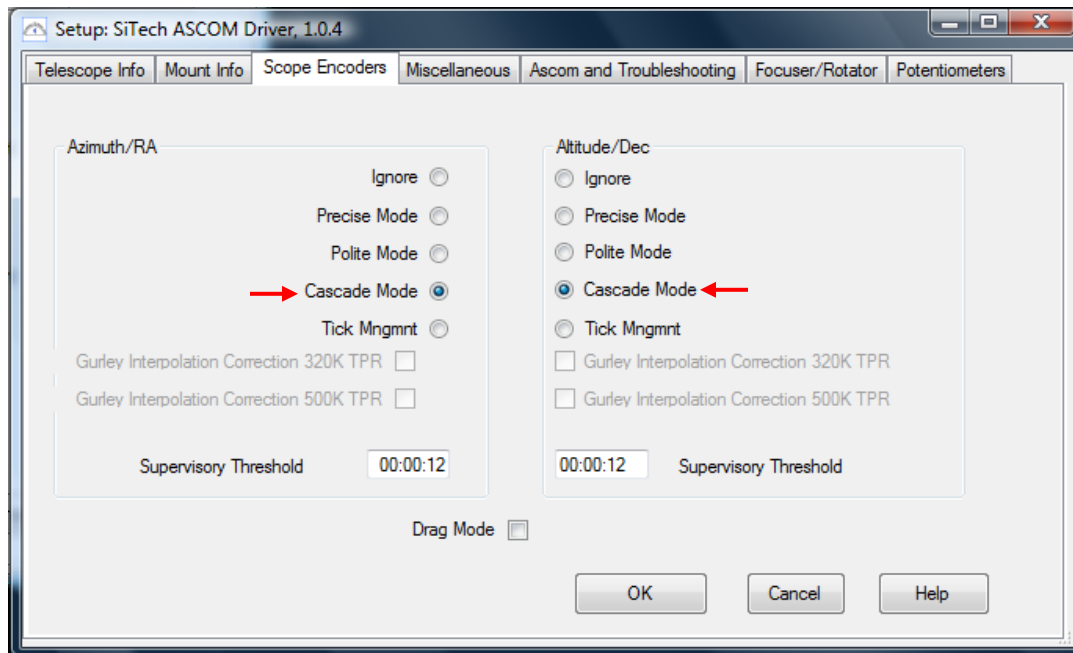
If you do not know your field of view, enter the Camera X Pixels, Camera Y Pixels, X Microns per Pixel, and Y Microns per Pixel (typically available from your camera spec sheet) and click "Calculate Field of View". The field of view will be calculated using these values, as well as the Focal Length value entered on the Telescope Info page.

Mount Info Tab



The default settings for the A200 are shown above. It is typically not necessary to change these.

Scope Encoders Tab



The default settings for the A200 with High-Resolution Axis Encoders are shown above. For mounts with high resolution encoders, like the A200HR, Cascade mode needs to be selected. To disable the axis encoders, select Ignore.

Polar Alignment, First Mount Model and Setting the Home Switches

NOTE: If you are located in the southern hemisphere, please refer to Appendix H before continuing.

The Ascension 200 mount is a German Equatorial mount. This means that when the mount's polar axis is aligned to the Earth's rotational axis, the mount need only track in one axis to counteract the rotation of the Earth.

Polar alignment is the process of adjusting the polar axis of the mount (also known as the Right Ascension axis) to be aligned with Earth's rotational axis. For non-computerized telescopes this is typically done using a process called the Drift method, an iterative process that is time consuming and requires some expertise.

Since the Ascension 200 is a computerized mount with sophisticated star-matching and pointing model software, the process is much easier. The first step is to make a small mount model, in which you align the telescope to about 6 or 8 known stars. Using this information, the mount modeling software will calculate the polar misalignment of the mount.

At that point the mount can be manually adjusted in azimuth and in latitude/altitude by the amount reported by the software. This process will bring the mount closer to proper polar alignment.

Steps before making your first mount model

The mount and the telescope should be installed as described in previous chapters. After the initial installation, the mount should be within ± 5 degrees from true North in azimuth and within ± 5 degrees of the correct latitude.

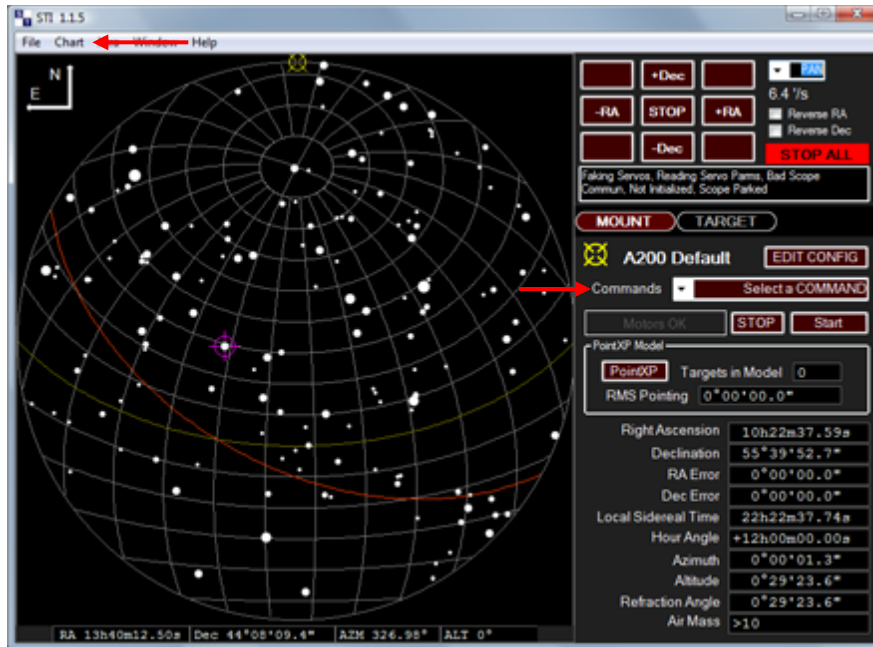
Before continuing, make sure that the telescope electronics are powered on, and that STI is properly connected to the mount.

You may also want to align a finder scope to the main imaging telescope before continuing.

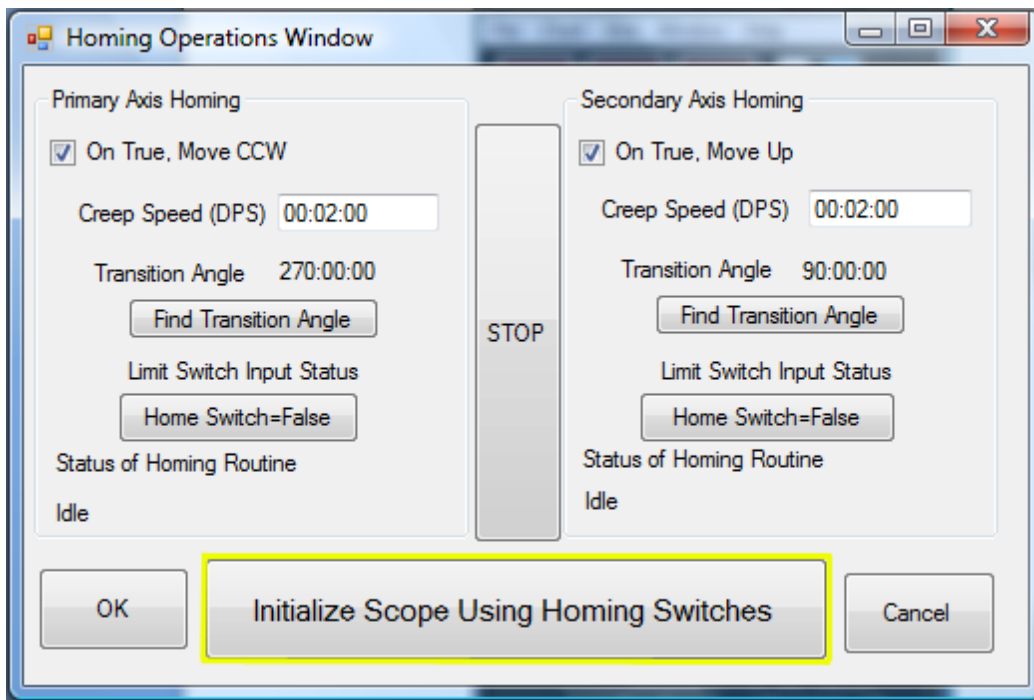
Creating a Mount Model using PlateSolveXP

In order to determine the mount's polar misalignment, you will need to create a small mount model. Since the telescope is not polar aligned, the pointing accuracy of the mount will initially be very poor as you move around to different points in the sky. We will start by making a small mount model using an aligned finder scope. We will then make a more accurate automated mount model using the main imaging telescope and CCD camera to precisely determine the mount's polar misalignment.

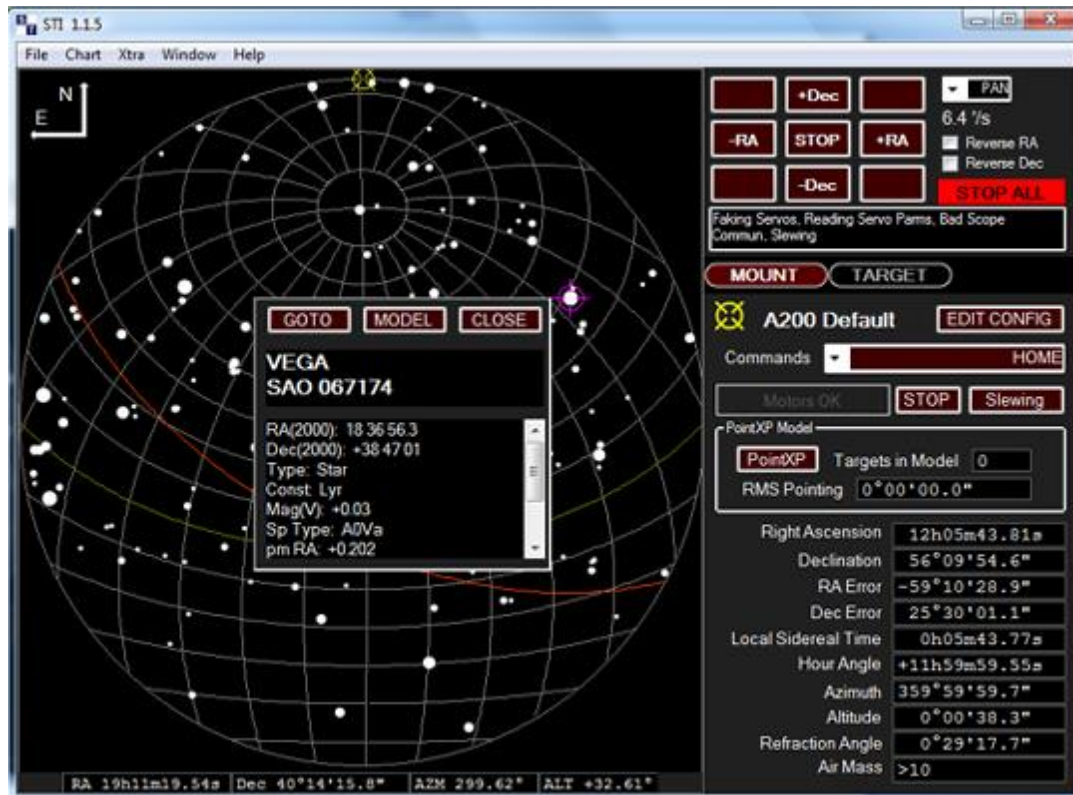
1. Open STI
2. Open Maxim DL and connect to your CCD camera. If you have a filter wheel, select the "brightest" filter -- that is, the filter that allows the most light through (typically Luminance, Clear, or Empty).
3. Select the Chart menu from the top menu bar of the STI window. Select "Show Sky Chart". This will open the sky chart window.



4. Click on the "Commands" menu and select "Home". The Homing Operations Window will open. Press the "Initialize Scope Using Homing Switches" button. The telescope will slew to its home position. For the A200, this corresponds to the orientation in which the counterweight bar is pointed down and the telescope is pointed at the celestial pole. This procedure, called "homing the mount", is used to calibrate the mount axes to a known position at startup.



5. If you had previously built a PointXP model (i.e. if "Targets in Model" is not zero), clear the old model by clicking the PointXP button, clicking "Clear Terms and Cal Stars", and then clicking OK.
6. Select a bright, easily identified star on the sky chart. A window will pop up telling you the name of the star. Click the GOTO button and the telescope will move approximately to the star you selected. It is not expected that you will be perfectly centered on the star since the telescope is not polar aligned, a mount model has not been established, and the home switches have not yet been trained.



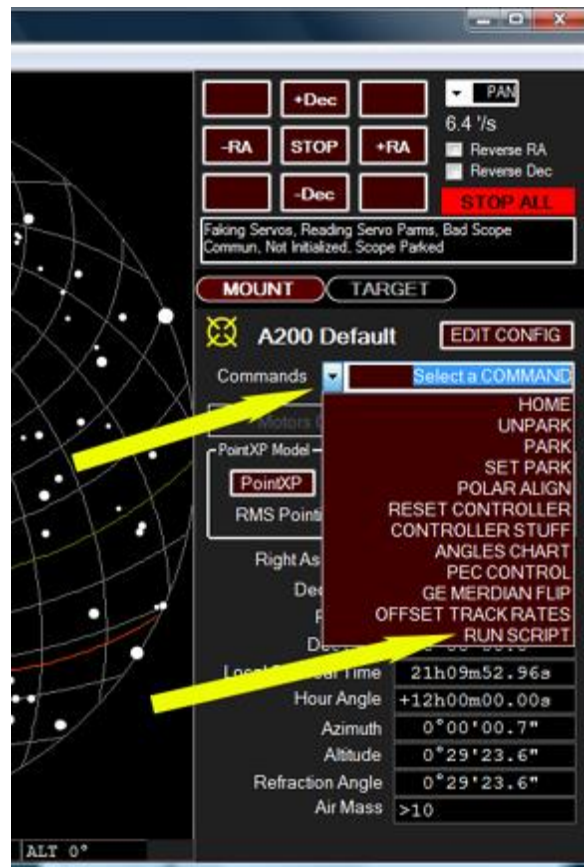
- Center the star in the finder scope using the hand paddle. Don't bother centering the star perfectly at this point. Press the MODEL button in the Sky Chart window, and the InitPoint window will open. It will assume you are going to do a Simple Offset Init (also known as a "Sync") shown by the yellow arrow below. There will be a 5 second count-down in case you wanted to do something else. A Simple Offset Init will offset the coordinates reported by the telescope so that they perfectly match the star that you have centered.

The screenshot shows the 'InitPoint' window with the following fields and buttons:

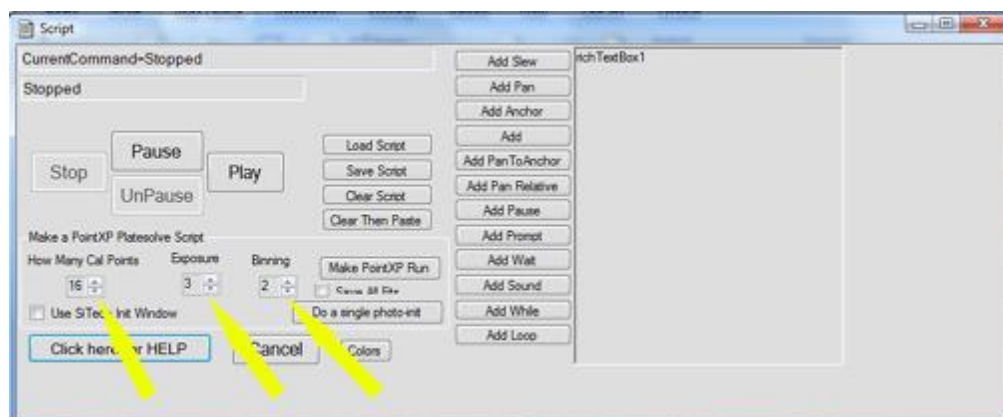
- Init Right Asc**: 02:04:35, 03:30:27
- Init Declination**: 42:23:11, 33:50:44
- Init Azimuth**: 297:51:01, -118:18:40
- Init Altitude**: 48:26:20, AltOffset
- Init Hour Angle**: 0, RMS=00:00:00
- Scope Latitude**: Peak=00:00:00
- Scope Longitude**: Just Load Star (No Terms)
- Altitude Offset**: Calc Only Offsets/Align
- Buttons**: Keep Window Up! (Manual Control), Simple Offset Init (automatic in 1) (highlighted with a yellow arrow), Load Cal Star No. 1, E-STOP, OK, Cancel, Help, Colors.
- Radio Buttons**: GFM Scope is looking West (selected), GFM Scope is looking Fast.

- The InitPoint window will close. Now press the MODEL button again. This time when the InitPoint window pops up, click the "Load Cal Star" button. This action will add the first calibration point to the PointXP pointing model. Telescope pointing will gradually improve as more calibration points are added to PointXP, and we will eventually use PointXP to determine the polar axis misalignment.
- Go back to the Sky Chart, choose another bright star, and press GOTO. Center the star in the finder scope and press MODEL. When MODEL is pressed and the InitPoint window opens, be sure to press the Load Cal Star Button. Repeat this process so you have 3 stars on one side of the meridian and 3 stars on the other side of the meridian.

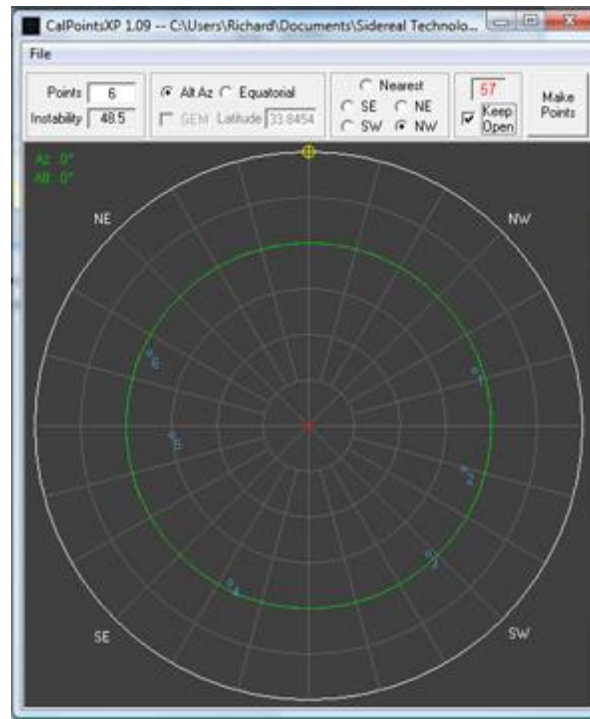
10. Now it is time to build a more accurate mount model using the CCD camera on your telescope. Go to the Commands menu and select Run Script at the bottom.



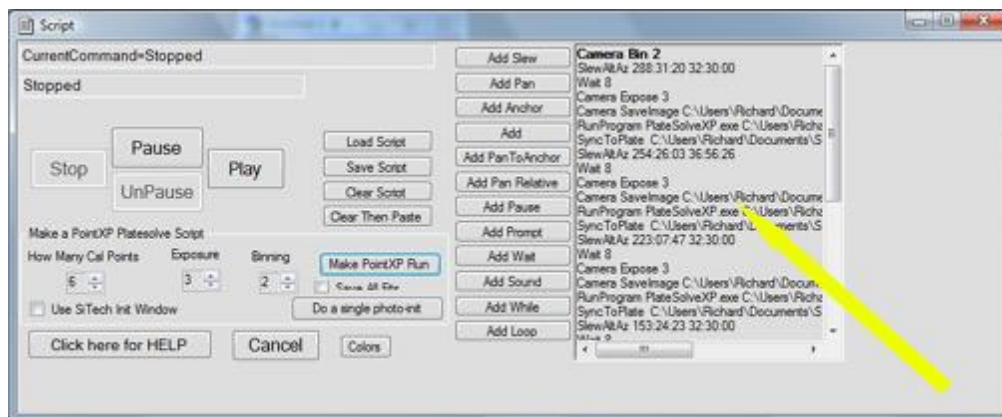
11. The Script window will open as seen below. The three arrows point to the most important settings for doing an automatic mount model run. For this initial polar alignment step, you should change “How Many Cal Points”, to 6. Later, when doing a full pointing model, you will want to map 30-40 points. Exposure time depends on your telescope, camera, selected filter, and site conditions. It should be long enough to get at least 10 good star detections that can be used for an automatic PlateSolve. Binning should normally be set to 2.



12. Once the parameters described above are set, press the “Make PointXP Run” button. This will open a new program called CalPointsXP which will select the optimum positions for the cal points. The window will automatically close after 60 seconds, but you can simply close it immediately before proceeding.

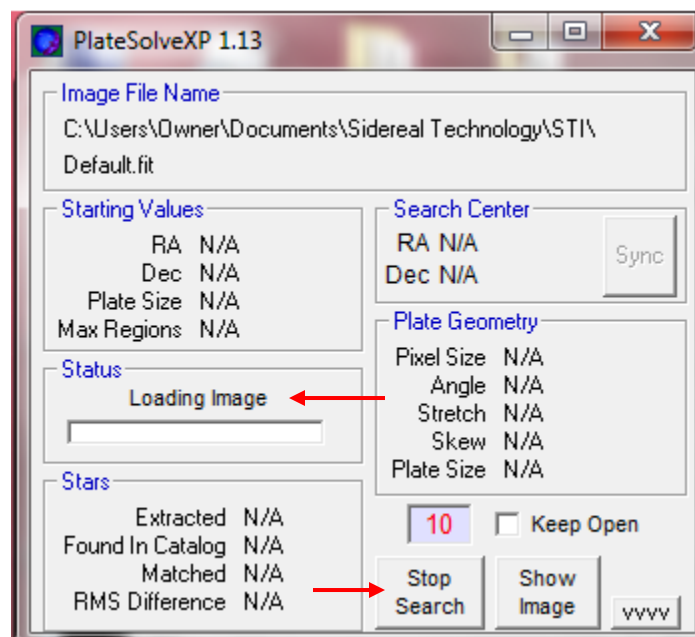


You should now see a pointing run script in the Script window, shown with the yellow arrow below.

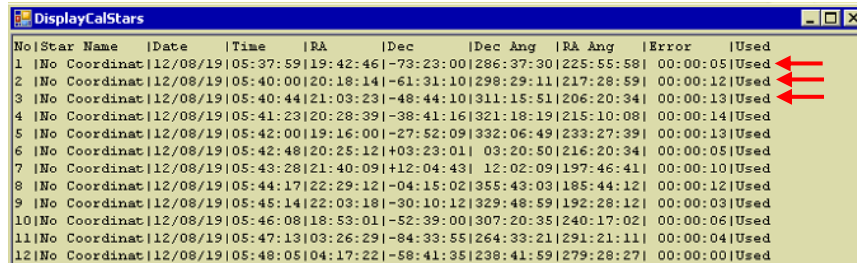


13. Press the PLAY button in the Script window. This will run the script and start the auto mapping run. The PointXP script will cause the telescope to slew to the first calibration point position, take an image, plate solve the image (accurately determine the position of the telescope by comparing the stars in the image to a known star catalog), and add the position information to the mount model.

The window that pops up is the PlateSolveXP window. The "Status" section will show the progress of the plate matching process, which involves loading the image, identifying ("extracting") stars in the image, and searching in a star catalog around the region of the mount's reported coordinates to find a match. PlateSolveXP will search up to 999 regions in an expanding spiral pattern before giving up. Once you have a few points in your pointing model, you should typically get a match after searching fewer than 50 regions. If you want to abort a search on a poor-quality image, click "Stop Search", and STI will proceed to the next calibration point.

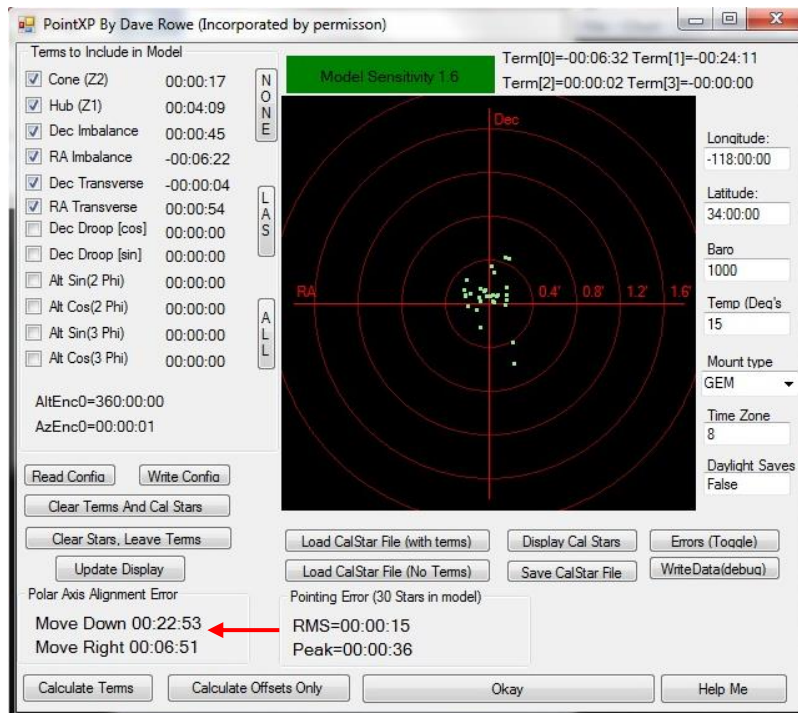


14. After the mount model is complete, press the PointXP button on STI. The PointXP window will pop up. (Note that the example screenshot below shows a much larger model than you will currently have).
15. Click "Display Cal Stars". This will show a list of the calibration points in the model. Remember that the first several calibration points were added using imperfect centering with a finder scope, and we don't want these points to pollute the model. Click on the "Used" text on the right column to remove these points from the model. If you took 6 points with the finderscope, then you should disable the first 6 points in the model, leaving only the highly accurate PlateSolve-based calibrations in the model. Close the window when finished.



No	Star Name	Date	Time	RA	Dec	Dec Ang	RA Ang	Error	Used
1	No Coordinat	12/08/19	05:37:59	19:42:46	-73:23:00	286:37:30	225:55:58	00:00:05	Used
2	No Coordinat	12/08/19	05:40:00	20:18:14	-61:31:10	298:29:11	217:28:59	00:00:12	Used
3	No Coordinat	12/08/19	05:40:44	21:03:23	-48:44:10	311:15:51	206:20:34	00:00:13	Used
4	No Coordinat	12/08/19	05:41:23	20:28:39	-38:41:16	321:18:19	215:10:08	00:00:14	Used
5	No Coordinat	12/08/19	05:42:00	19:16:00	-27:52:09	332:06:49	233:27:39	00:00:13	Used
6	No Coordinat	12/08/19	05:42:48	20:25:12	+03:23:01	03:20:50	216:20:34	00:00:05	Used
7	No Coordinat	12/08/19	05:43:28	21:40:09	+12:04:43	12:02:09	197:46:41	00:00:10	Used
8	No Coordinat	12/08/19	05:44:17	22:29:12	-04:15:02	355:43:03	185:44:12	00:00:12	Used
9	No Coordinat	12/08/19	05:45:14	22:03:18	-30:10:12	329:48:59	192:28:12	00:00:03	Used
10	No Coordinat	12/08/19	05:46:08	18:53:01	-52:39:00	307:20:35	240:17:02	00:00:06	Used
11	No Coordinat	12/08/19	05:47:13	03:26:29	-84:33:55	264:33:21	291:21:11	00:00:04	Used
12	No Coordinat	12/08/19	05:48:05	04:17:22	-58:41:35	238:41:59	279:28:27	00:00:00	Used

16. In the lower left corner you will see the window labeled "Polar Axis Alignment Error"



In the Polar Alignment Error window, you will see the error is listed as "Move Down dd:mm:ss" (degrees, arcminutes and arcseconds) and "Move Right dd:mm:ss". This tells you how far you will need to lower the polar axis in altitude, and rotate the polar axis in azimuth. In the case of the window above, the polar axis should be lowered 22 arcminutes and 53 arcseconds, and moved to the right 6 arcminutes and 51 arcseconds.

17. To move the polar axis in azimuth, loosen the six azimuth locking knobs shown in Figure 26 below. Assuming the polar axis needs to be moved to the right, first loosen the azimuth adjustment knob on the left, then tighten the Azimuth Adjustment Knob on the right.

NOTE 1: For the Azimuth Adjustment Knob: 1 turn = 0.4 degrees (24 arcminutes).

NOTE 2: For a Mathis mount, please see the included mount manual for this information.

In the example above, the polar alignment needs to be adjusted by 6 arcminutes and 51 arcseconds. That is the same as 6.85 arcminutes. See Appendix D for the conversion equation.

To calculate how much to turn the azimuth adjustment knobs, take the polar misalignment in arcminutes and divide it by 24. This will give you the number of turns. In our example,

$6.85/24 = .28$ turns, or just over a $\frac{1}{4}$ turn. We would loosen the azimuth adjustment knob on the left by at least $\frac{1}{4}$ turn, tighten the azimuth adjustment knob on the right by $\frac{1}{4}$ turn (which rotates the entire mount in azimuth), and then tighten the left knob.

Once the azimuth has been adjusted, tighten the six azimuth lock knobs.

To move the polar axis in altitude, first loosen the Altitude Lock knobs. In our example, the polar axis needs move down 22 arcminutes and 53 arcseconds, or 22.88 arcminutes. (See Appendix D)

NOTE: For the Altitude Adjustment Knob: 1 turn = $\frac{1}{2}$ degree (30 arcminutes)

To calculate how many turns to move the Altitude Adjustment Knobs, take the polar misalignment in arcminutes and divide it by 30. This will give you the number of turns. In our example,

$22.88/30 = 0.95$ turns, or just about 1 full turn.

Once the altitude has been adjusted, tighten the Altitude Lock knobs.

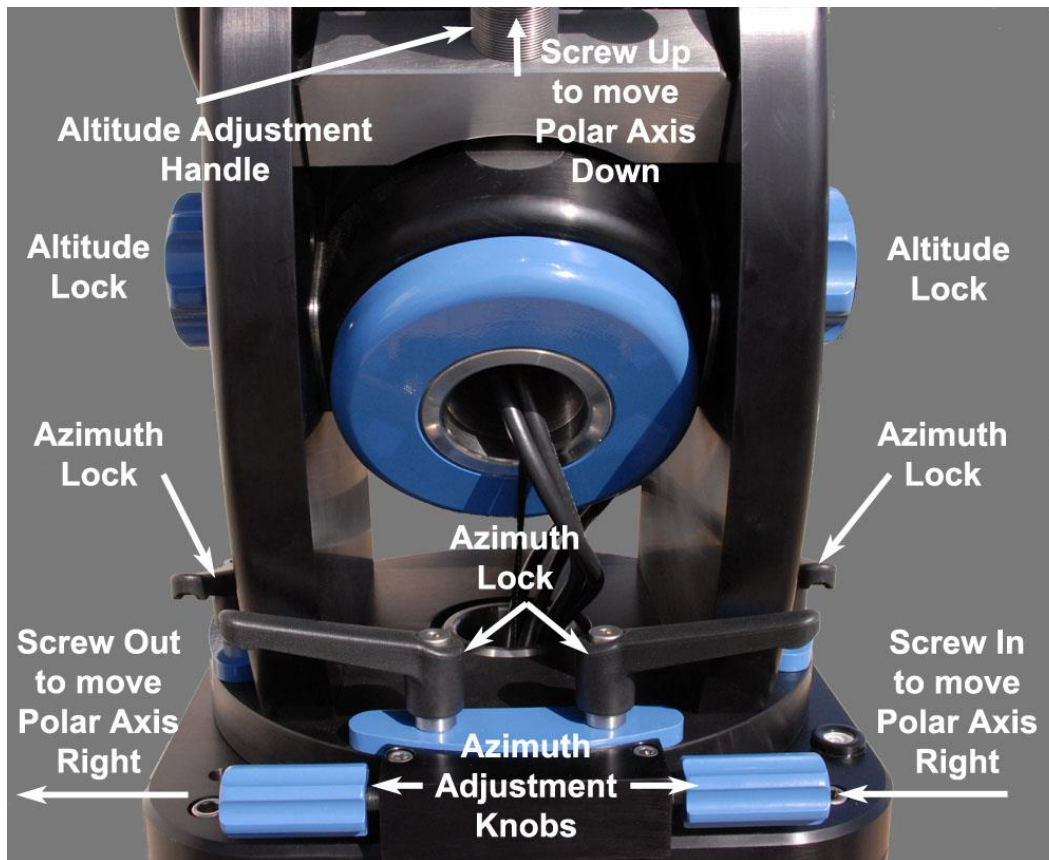


Figure 26 - Polar alignment adjustment components

Your mount should now be much closer to proper polar alignment. A new mount model should be created, as described in the next section. If the new mount model still indicates a significant polar misalignment, repeat these steps until the alignment error is under about 10 arcminutes in each axis. The best you can expect to do with reasonable effort is about 1 arcminute of alignment error in each axis. The PointXP model will be able to compensate for any residual misalignment.

Mount Model

A mount model is a very accurate way of aligning the telescope to the sky. It improves both the pointing and the tracking of the telescope system. It does this by comparing a series of mount positions against positions on the sky, measured by either performing a PlateSolve or manually centering a star in the telescope. The model has several different correction variables, and these variables are fit against the measured position errors to minimize the RMS error and improve telescope pointing. A reasonable model should contain 16-40 points distributed across the sky. However, the mount should begin pointing reasonably well after just 4 or 5 points.

Once a mount model is created, it can be saved and reused each time the telescope is powered on and homed. The model should remain valid until some physical alignment of the mount, telescope, or imaging system changes.

When the mount is first powered on, it is "lost" and needs to be calibrated against some known reference position. There are three common ways of initializing the mount:

1. **"Homing" the mount.** This instructs the mount to slew to a position where a set of "home switches" are triggered. These switches are located at a known position, and can be used to initialize the coordinate system of the mount.
2. **"Parking" and "Unparking" the mount.** When the mount is in a known position, "Parking" the telescope will cause it to slew to a predefined position and then stop all motion. The mount can then be turned off, and turned back on at a later time. When the mount is "Unparked", the control system assumes that the mount is still positioned at the same point as when it was parked, and initializes the coordinate system using that information. (**Note:** If the mount has been moved, e.g. by loosening and tightening the clutch knobs, since the last Park, the mount will not be correctly positioned for an Unpark and must be initialized using one of the other methods.)
3. **Calibrating to a sky position using Offset Init.** At startup, the mount is manually positioned to some reference point in the sky, such as a bright star. When the star is centered, an "Offset Init" is performed, which initializes the mount's coordinate system so that its current position corresponds to the current position of the target star. This is also known as "syncing" the mount.

The procedure for saving and associating a pointing model for each of these initialization techniques will be described below.

Things That May Invalidate the Mount Model

- Changing the collimation of the telescope
- Changing from a camera to an eyepiece with a diagonal, where the diagonal is not perfectly at 45 degrees.
- Changing telescopes – if you change telescopes, or remove and reinstall your existing telescope from the mount, you may need to create a new model
- Adding or removing significant weight from the telescope.

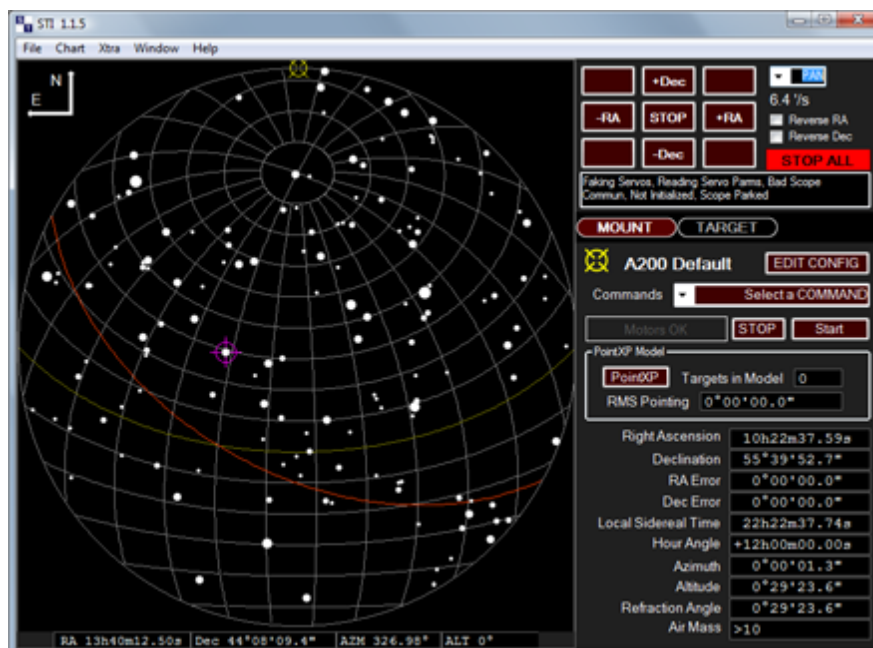
All of these changes will affect the mount model, although in some cases the effect may be small enough to ignore. To evaluate how well the telescope is pointing, try slewing to a few bright stars. For each bright star, take an image and measure how close to the center of the CCD the star is positioned. If the pointing accuracy is negatively impacted enough, you can create another mount model. You can even have multiple mount models, one for each equipment setup if you need it. Contact PlaneWave Instruments for details on how to do this.

Creating a Mount Model

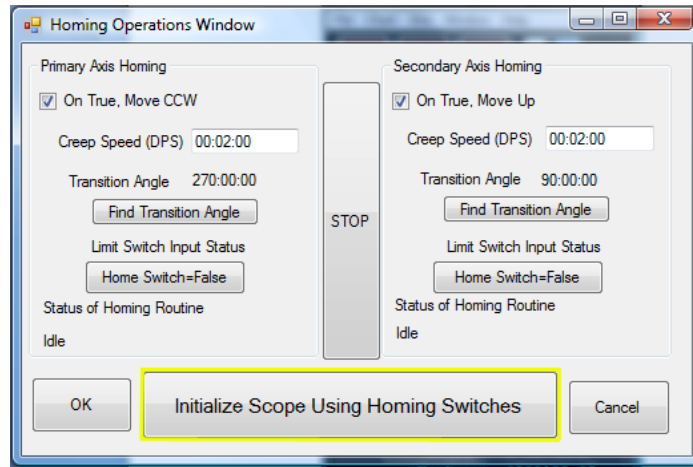
1. Open STI, if it is not already running.



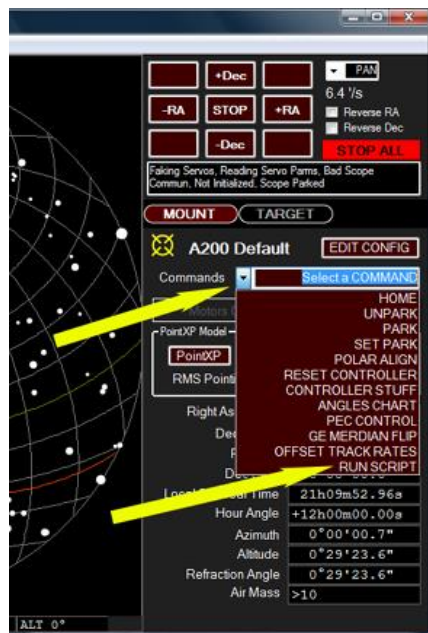
2. Open Maxim DL and connect to your CCD camera. If you have a filter wheel, select the "brightest" filter -- that is, the filter that allows the most light through (typically Luminance, Clear, or Empty).
3. Select the Chart menu from the list of options on the top of the STI window. Select "Show Sky Chart". This will open the sky chart window.



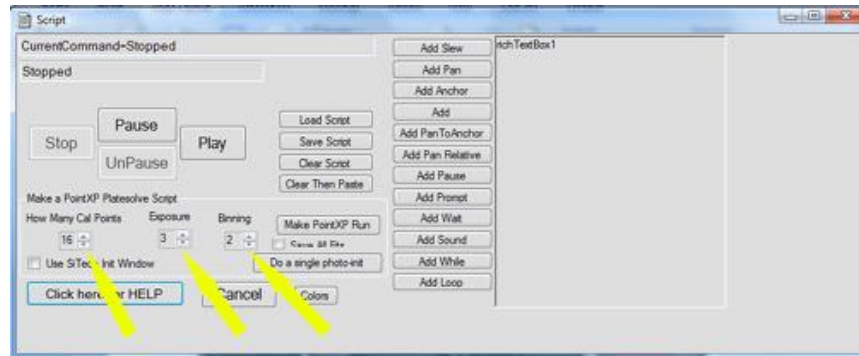
- Click on the "Commands" menu and select "Home". The Homing Operations Window will open. Press the "Initialize Scope Using Homing Switches" button. The telescope will slew to its home position. For the A200, this corresponds to the orientation in which the counterweight bar is pointed down and the telescope is pointed at the celestial pole.



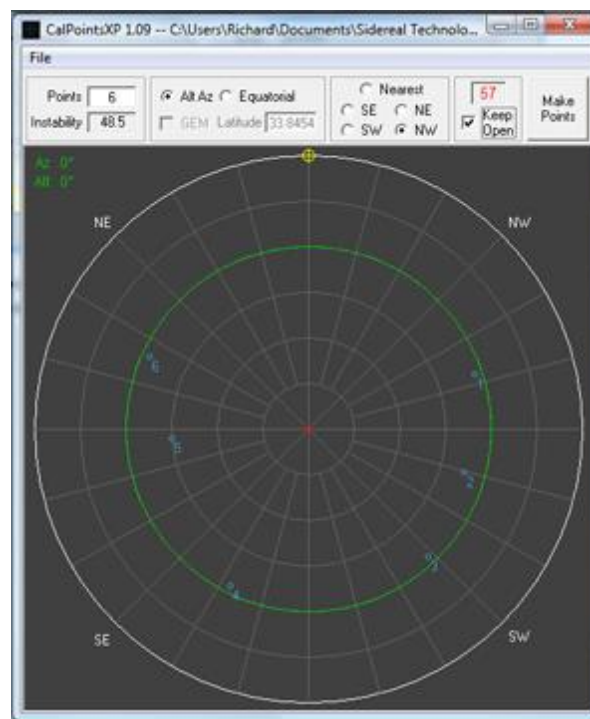
- Clear the old PointXP model by clicking the PointXP button, clicking "Clear Terms and Cal Stars", and then clicking OK.
- From the Commands menu, select Run Script.



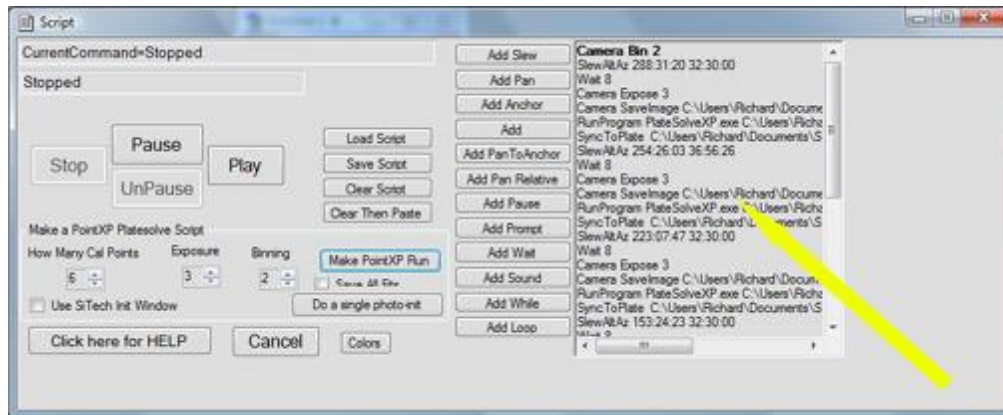
- The Script window will open as seen below. The three arrows point to the most important settings for doing an automapping run. Since this is going to be the main model for the telescope, set the number of stars to be somewhere between 16 and 40. More points will take longer to acquire, but should result in a more representative mount mode. Set the exposure time so that a typical image will result in at least 10 good star detections in an image. A binning value of 2 is normally OK, but if your stars are undersampled (smaller than 2x2 pixels in the resulting images), change this to 1.



- Press the “Make PointXP Run” button. This will open a new program called CalPointsXP which will select the optimum positions for the calibration points. You may close the CalPointsXP window immediately.

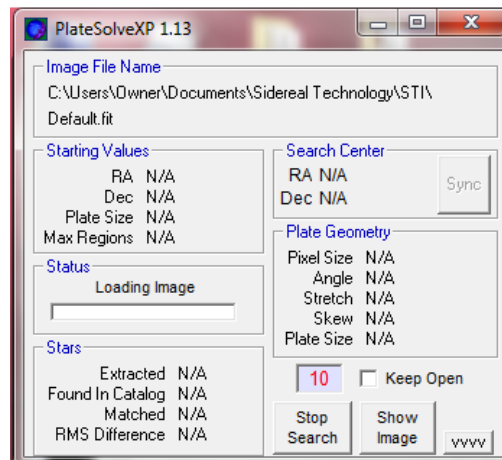


9. An automapping script will be generated in the scripting window, as shown below.



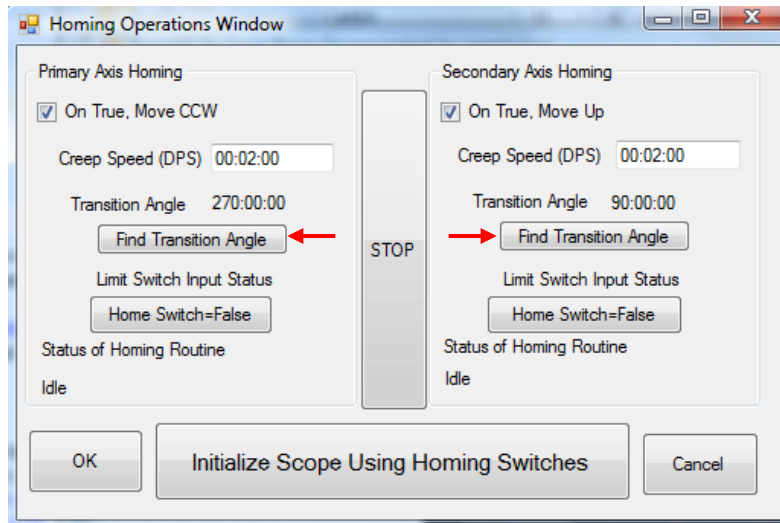
10. Press the PLAY button in the Script window. This will run the script and start the auto mapping run. The PointXP script will cause the telescope to slew to the first calibration point position, take an image, plate solve the image (accurately determine the position of the telescope by comparing the stars in the image to a known star catalog), and add the position information to the mount model.

The window that pops up is the PlateSolveXP window. The "Status" section will show the progress of the plate matching process, which involves loading the image, identifying ("extracting") stars in the image, and searching in a star catalog around the region of the mount's reported coordinates to find a match. PlateSolveXP will search up to 999 regions in an expanding spiral pattern before giving up. Once you have a few points in your pointing model, you should typically get a match after searching fewer than 50 regions. If you want to abort a search on a poor-quality image, click "Stop Search", and STI will proceed to the next calibration point.



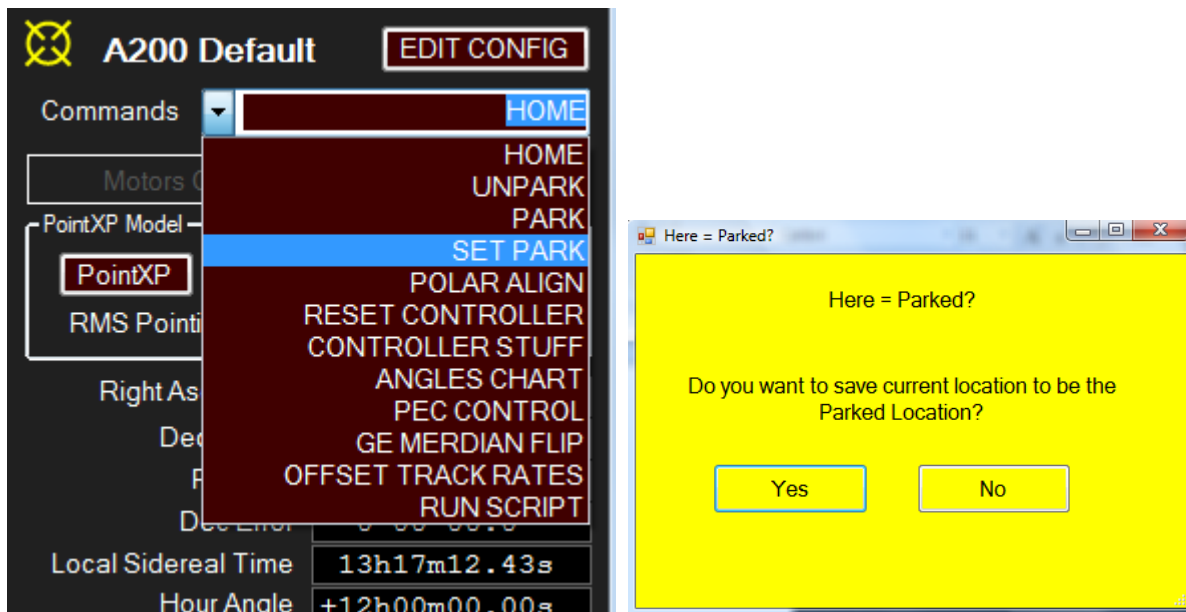
11. After the mount model is complete, press the PointXP button on STI and the PointXP window will pop up. You may inspect the model, and reject points that appear to be erroneous.

12. Now that you have a complete mount model, you must calibrate the positions of the home switches using the model information. You must do this one axis at a time. Under Commands list, select HOME. This will open the HOME dialog box as seen below.



13. Notice that there are two Find Transition Angle buttons on the dialog box, one for the Primary Axis (RA) and one for the Secondary Axis (DEC). Press the Find Transition Angle button for the Secondary Axis first. For the A200, the mount will slew in declination until the telescope is pointed at the pole. A new number for the Transition Angle will be displayed that is close to, but not exactly, 90 degrees. After the Secondary Axis has been calibrated, press the Find Transition Angle for the Primary Axis. For the A200, the mount will slew in right ascension until the counterweight bar is pointed down. When it is complete, the transition angle should be close to, but not exactly, 270 degrees.
14. Press the OK button. This will save the PointXP file as HomePointXP.PXP for future use, and will also save the new home switch transition angles. **Do not click cancel or simply close the Homing Operations window, or this information will be lost and you will need to build a new pointing model!**
15. If desired, set a park position. Slew the telescope manually with the hand pad or the buttons on STI until the telescope is at the desired park position. For roll-off observatories, this usually involves pointing the telescope in a horizontal position so as to clear the roll off roof.

16. Once the telescope is at the desired park position, hit the STOP button to turn off tracking. Then from the Commands menu, select SET PARK. Once this is done, open the Commands menu and select PARK. This writes the current mount model to a file called ParkPointXP.PXP for future use.



Using the Telescope After Creating a Mount Model

Startup and shutdown of the Mount

Once you have polar aligned and built a PointXP model for the mount, it is ready for general use. Normally the telescope will be used in the evening and shut off during the day. This section talks about how to start up and shut down the mount.

There are three ways to get up and running with an existing mount model. First, if you PARKed the telescope the last time you shut down the mount, you can UNPARK the telescope to retrieve its previous position. Once the mount has been unparked, the "Park" PointXP model will be loaded and the telescope will be ready to slew.

Another method to start up is to initialize the telescope using its home sensors. This assumes that you calibrated the home sensors the last time you build a pointing model. If you open the Commands menu and select the HOME option, the mount will move to its home position, initialize the coordinate system based on the calibrated home switch angles, and load the "Home" PointXP model.

The third option is to recall a previously saved mount model and then perform a Simple Offset Init (also called a "sync") on a reference star.

At the end of an observing session, you can PARK the mount, which will slew the telescope to the position you previously set as the PARK position, and turn off tracking. You can also simply turn off power to the mount, although if you do this you MUST home the mount the next time you turn it on. (You cannot use the UNPARK procedure). Finally, you can simply stop tracking and leave the mount powered on until your next observing session.

Control via external software

The STI control software manages the basic operation of the mount. However, STI does not have an extensive database of interesting targets in the sky beyond some bright stars and Messier objects. It also does not manage extended observing sessions for you. To support these operations, STI may be controlled from other programs using its ASCOM interface. It is very common to operate STI via another program, such as TheSky (see Appendix G), ACP, or CCDAutoPilot.

Autoguiding

Autoguiding is the process of taking repeated images of a guide star, detecting any drift or positioning error on that star, and sending a correction offset to the mount to put the star back at the desired position.

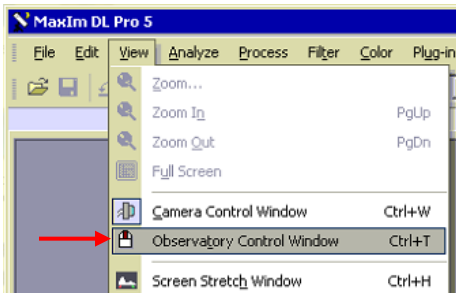
There are two different ways to send these offset signals. One way is called "relay guiding". This is where the guide camera is connected to the motor control box and as the guide star drifts, the CCD camera sends electrical guide signals to the motor control box. In a sense, the guide camera is pressing virtual handpaddle buttons to nudge the mount's position slightly. There is a port labeled "Autoguider" on the motor control box for this use. However, relay guiding bypasses the axis encoders on the A200 and therefore does not produce a highly accurate offset. **Relay guiding is not recommended for the A200 mount or any other mount with high resolution encoders.**

The recommended method of autoguiding for the A200 is called "pulse guiding", also known as "ASCOM guiding". In this method, as the guide star drifts on the guide camera, the camera control software (for example, MaxIm DL)

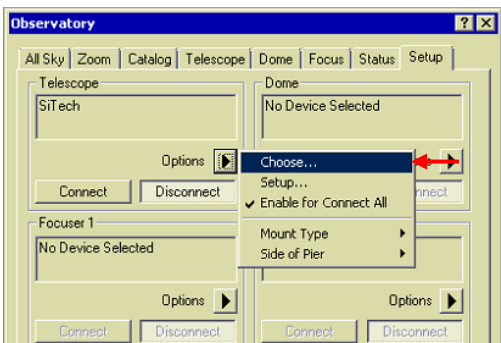
notes this movement and sends ASCOM commands to STI in order to make the appropriate corrections. Pulse guiding is beneficial because it doesn't use any additional cables and it uses feedback from the high resolution axis encoders to produce extremely accurate offsets.

To set up pulse guiding with MaxIm DL version 5:

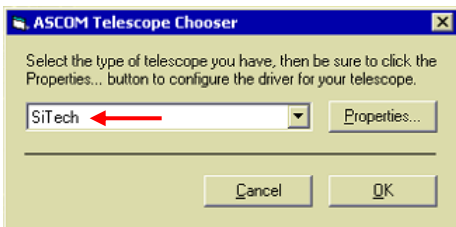
1. From the "View" menu, open the "Observatory Control Window"



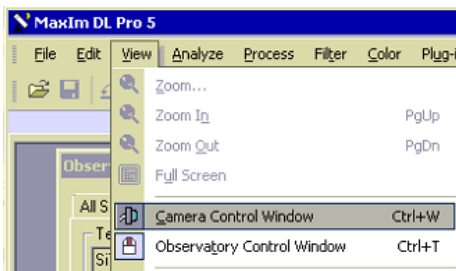
2. In the "Observatory" window, under the "Setup" tab in the "Telescope" section, open the "Options" menu and select "Choose..."



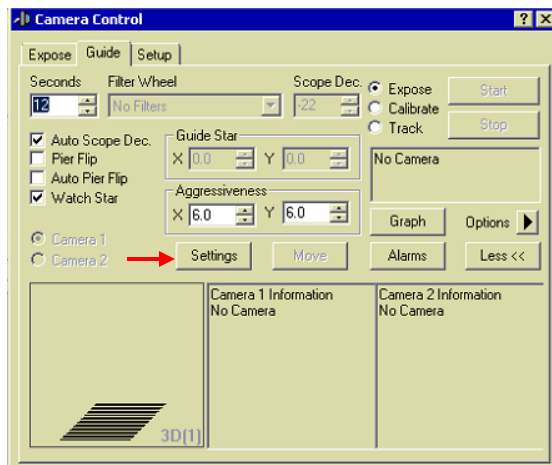
3. In the "ASCOM Telescope Chooser" window, select "SiTech" or "PlaneWave STI" and click "OK".



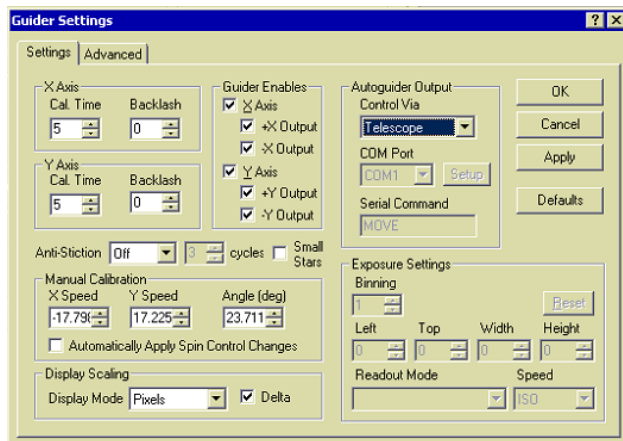
4. In the "Observatory" window under the "Telescope" section, click "Connect"
5. In the main MaxIm DL window, open the "View" menu and select "Camera Control Window"



6. In the "Camera Control" window, select the "Guide" tab and click "Settings"



7. In the "Guider Settings" window under "Autoguider Output", pull down the "Control Via" menu and select "Telescope". Click "OK".



MaxIm will now send all guide commands via ASCOM to the configured Telescope device -- in this case, STI.